

Class 5: Linear Independent Sets

A set of vectors $\{v_1, v_2, \dots, v_k\}$ is linearly independent if $\vdots$

In other words

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If there is some non trivial linear combination of the vectors that produces the 0-vector we say the set of vectors is _____.

Example: Are the vectors $\begin{bmatrix} 1 \\ 0 \\ -2 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ -7 \\ 4 \end{bmatrix}, \begin{bmatrix} -2 \\ -6 \\ 7 \\ 9 \end{bmatrix}$ linearly independent?

How To:

How do you decide if a set of vectors is linearly independent?

1.

2.

3.

Three shortcuts

1. Any set containing the 0-vector is linearly **dependent**.
2. Any set of more than n vectors in $\mathbf{R}^n$ will be linearly dependent.
3. A set of two vectors will be linearly dependent if and only if one vector is a scalar multiple of the other one.